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$$T_4(f, 0)(x) \text{ voor } f(x) = \frac{1}{1+x}$$

$$T_n(f, a)(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

$$f'(x) = \left(\frac{1}{1+x} \right)' = \frac{1}{(1+x)^2}$$

$$\Rightarrow f'(0) = -1$$

$$f''(x) = \left(\frac{1}{1+x} \right)'' = \frac{-2}{(1+x)^3}$$

$$\Rightarrow f''(0) = 2$$

$$f'''(x) = \left(\frac{1}{1+x} \right)''' = \frac{6}{(1+x)^4}$$

$$\Rightarrow f'''(0) = -6$$

$$f^{iv}(x) = \left(\frac{1}{1+x} \right)'''' = \frac{-24}{(1+x)^5}$$

$$\Rightarrow f^{iv}(0) = 24$$

$$T_4(f, 0)(x) = 1 - x + \frac{2x^2}{2!} - \frac{6x^3}{3!} + \frac{24x^4}{4!}$$

$$T_4(f, 0)(x) = 1 - x + x^2 - x^3 + x^4$$

References